AI Use Zones for MEC4126F

A Framework for Responsible Generative AI Use

UCT Framework for AI in Education

In June 2025, UCT introduced the UCT Framework for Artificial Intelligence in Education with three principles:

1. **Promote responsible and ethical AI literacies for staff and students**
2. **Ensure academic integrity**
3. **Enhance teaching and learning**

Policy available at: UCT AI in Education Framework

The MEC4126F AI Use Zones are aligned with this foundation to help ensure:

- AI tools **enhance** rather than replace learning.
- **Academic integrity** is maintained across all assessments.
- Critical **skills required** of professional engineers are developed.

Introduction to AI Use Zones

The use of generative AI tools in education presents both opportunities and challenges.

This framework establishes three zones to help understand how AI can be used responsibly:

- 🟢 **Green Zone**: AI as a Learning Tool
- 🟡 **Yellow Zone**: AI-Assisted Work with Full Understanding
- 🔴 **Red Zone**: Prohibited Use

Goal: Support learning while maintaining academic integrity and developing required engineering skills.

Green Zone: AI as a Learning Tool

AI acts as a tutor, debugger, and study aid; **not as a replacement** for thinking.

Appropriate Uses:

1. **Concept Explanation**: Explaining concepts, syntax, or errors.
2. **Debugging Assistance**: AI assists with identifying problems in your code.
3. **Code Review**: Getting feedback to improve style, efficiency, or clarity.
4. **Documentation Help**: Understanding library functions or registers.
5. **Study Aid**: Generating practice problems.
6. **Time Management**: Assisting with planning your time.
7. **Grammar and Syntax**: Using an AI to help you write with more correct grammar or to identify how you haven't adhered to the programming language requirements.

Green Zone: Example & Key Principle

Interaction Example:

> *Student:* "I'm getting a build error in code I wrote. I've read the debug trace but can't figure it out. Can you help?"
> *AI:* [Explains likely cause and suggests solution]
> *Student:* [Uses explanation to fix code and learns]

Key Principle:

AI is used as a **learning tool**. *If a student cannot explain their solution line-by-line, they have not used AI appropriately.*

🟡 Yellow Zone: AI-Assisted Work

AI-generated code may be incorporated under **strict conditions** that ensure learning still occurs.

Conditions for Use:

1. **Full Comprehension**: You must be able to explain, modify, and debug every line independently.
2. **Substantial Original Work**: The majority of the codebase must be your own.
3. **Testing and Verification**: AI code must be thoroughly tested in context.
4. **Documentation**: You must document which parts were AI-assisted.
5. **No Core Algorithms**: Do not use AI for primary algorithmic/design logic being tested.

🟡 Yellow Zone: Documentation Requirements

If using AI in the Yellow Zone, include a brief note in your submission:

```
AI Use Declaration:
- Used AI to generate the moving average filter function (lines 45-62).
- Modified the function to use fixed-point arithmetic for performance.
- Verified correctness against known test cases.
- Fully understand the implementation and can explain it.
```

Key Principle:

AI is used as a **technical tool**, but is not a replacement for expertise. A student remains fully responsible for the final submission. Transparency is **not** penalised and is encouraged.

Red Zone: Prohibited Use

Uses that **undermine learning** objectives and may constitute academic misconduct.

Prohibited Uses:

1. **Direct Submission**: Submitting without understanding, modifying, or verifying.
2. **Complete Solutions**: Asking AI to solve entire problems.
3. **Obfuscation**: Modifying code to hide that it was AI-generated.
4. **Exam/Test Use**: Using AI during closed-book assessments.
5. **False Attribution**: Claiming AI work as your own.
6. **Bypassing Learning**: Using AI without engaging with concepts.

🔴 Red Zone: Example of Prohibited Use

Interaction Example:

> *Student:* "Write me a complete STM32 program that reads an ADC, applies a moving average..."
> *Student:* [Submits code without understanding, cannot explain during oral assessment]

Key Principle:

AI is doing the thinking, **learning is undermined**. Submitting work from the Red Zone will be treated as a violation of academic integrity.

Guidelines for Responsible AI Use

Before using AI generated work, ask:

1. *Have I attempted this myself first?*
   AI use should help overcome sticking points, not to prevent them from occurring. These points are often where the most learning happens.
2. *Do I understand the result?*
   If you can't explain it, don't use it.
3. *What am I actually learning?*
   If "nothing", reconsider your approach.
4. *Could I do this in an exam?*
   If not, you need more practice without AI.

Guidelines for Responsible AI Use

> Responsible AI use is a skill. Build good habits:
>
> - **Verify everything**: AI can generate plausible nonsense (which can damage hardware).
> - **Understand limitations**: AI lacks context on your specific hardware unless told.
> - **Develop critical evaluation**: Learn to spot wrong, inefficient, or inappropriate output.
> - **Practice without AI**: Regularly solve problems independently to maintain skills.

Assessment-Specific Policies

Assessment Type	Typical AI Policy
Weekly practicals / Problem sets	**Green & Yellow Zone** permitted
In-lab practical exam	**No AI use** (simulates field conditions)
Tests and exams	**No AI use** (closed-book, closed-internet)

When in doubt, ask before submitting!

Summary of AI Use Zones

Zone	Description	Analogy
🟢 **Green**	AI as tutor/debugger. Full understanding required.	Using a calculator to check work.
🟡 **Yellow**	AI-assisted with transparency. You remain responsible.	Using CAD software for a part you understand.
🔴 **Red**	AI replaces thinking. Academic misconduct.	Submitting someone else's work as your own.

Thank You

Questions?